Question	Answer	Marks	Partial Marks	Guidance
6(a)	$\begin{pmatrix} 4 \\ -3 \end{pmatrix}$	**2**	**B1** for $\begin{pmatrix} 6 \\ -10 \end{pmatrix}$ or answer $\begin{pmatrix} 4 \\ k \end{pmatrix}$ or $\begin{pmatrix} k \\ -3 \end{pmatrix}$	Penalty 1 mark if fraction line in vector
6(b)(i)	(–4, –1)	**1**		
6(b)(ii)	7.21 or 7.211…	**2**	**M1** for $(-6)^2 + 4^2$	Condone -6^2 if recovery seen Accept $\sqrt{52}$ or $2\sqrt{13}$ as final answer
6(c)(i)(a)	2**c** – **a**	**1**		Accept e.g. –1**a** + 2**c**
6(c)(i)(b)	**c** – **a**	**1**		Accept e.g. –1**a** + 1**c**
6(c)(i)(c)	$\frac{1}{5}$(**a** + 8**c**) final answer	**2**	**M1** for [$\overrightarrow{AP}$ =] $\frac{4}{5}$ × *their*(2**c** – **a**) or [$\overrightarrow{BP}$=] $\frac{1}{5}$ × – *their* (2**c** – **a**) or for a correct vector route using the lines on the diagram	For 2 marks, accept e.g. $\frac{1}{5}$**a** + $\frac{8}{5}$**c** , 0.2**a** + 1.6**c** Condone mixed notation for vector route e.g. 2**c** + *BP*
6(c)(i)(d)	$\frac{4}{5}$(– **a** + **c**) final answer	**2**	**M1** for [$\overrightarrow{QP}$ =] – $\frac{4}{5}$**c** + $\frac{4}{5}$ × *their*(2**c** – **a**) or for a correct vector route	For 2 marks, accept e.g. – $\frac{4}{5}$**a** + $\frac{4}{5}$**c,** –0.8**a** + 0.8**c** Condone mixed notation for vector route - may use *their* (c)(i)(c) in this route e.g. *QA* + –**a** + *their OP*
6(c)(ii)	[*QP* is] parallel [to *CB*] *QP* = $\frac{4}{5}$*CB* oe	**2**	**Dep both statements** consistent with *their* (c)(i)(b) and *their* (c)(i)(d) and both vectors in terms of **a** and **c** **B1** for each dep on statement consistent with *their* (c)(i)(b) and *their* (c)(i)(d) and both vectors in terms of **a** and **c**	Accept e.g. *QP* ∥ *CB*, *QP* : *CB* = 4 : 5, $\overrightarrow{QP} = \frac{4}{5}\overrightarrow{CB}$ Do not accept contradictory statements e.g. they are parallel and perpendicular Do not accept statements if they are not consistent with *their* (c)(i)(b) and (c)(i)(d)